Data-Backed Journal-Style Replication of *Position and Differentiation of Firms in Technology Space*

Replication Paper Project

2026-04-04

Abstract

This paper presents a local calculated replication of Position and Differentiation of Firms in Technology Space using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 5,192 observations across 443 entities, and the batch headline estimate is -0.1935 for tech_diff with p = 0.7841. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: technology differentiation; firm performance; patent portfolios; local calculated replication

1 Introduction

This replication begins from a paper-first reading of Position and Differentiation of Firms in Technology Space and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether technologically differentiated firms perform better than their peers. The study is worth revisiting because The replication matters because it tests whether a central strategy result about technological differentiation sur-

vives when the semantic layer is stripped back to a fully executable local panel. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted. The executable evidence base used in this manuscript contains 5,192 observations across 443 entities and spans 1991 to 2015 in the local executable panel.. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets. The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions.

2 Original Study, Theory, and Replication Boundary

The focal source article is [1]. The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper builds technology-space measures from patent text and studies differentiation among U.S. public firms. The main outcome is firm performance, especially Tobin's Q. Technology differentiation is defined relative to technologically similar peers in the same year. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data. In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue. The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into

verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design.

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper uses semantic patent content. The local reconstruction therefore adopts the following bounded decision: The local rerun uses processed primary-USPC class portfolios. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Tobin's Q is a central outcome. The local reconstruction therefore adopts the following bounded decision: The local panel keeps Tobin's Q because it is fully reconstructible. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Differentiation is relative to peers. The local reconstruction therefore adopts the following bounded decision: Peer comparison is preserved with a class-based similarity measure. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline.

DirectSourceFact	LocalReplicationDecision
The source paper uses semantic patent content.	The local rerun uses processed primary-USPC class portfolios.
Tobin's Q is a central outcome.	The local panel keeps Tobin's Q because it is fully reconstructible.
Differentiation is relative to peers.	Peer comparison is preserved with a class-based similarity measure.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are DISCERN-linked firm-patent assignments with unambiguous GVKEYs. PatentView-derived utility-patent and primary-USPC class information. Compustat annual fundamentals used to reconstruct Tobin's Q and controls. Operationally, the reconstruction moves through the following file chain: Analysis panel is backed by the local file arts_2023_analysis_dataset.csv; Coefficient export is backed by the local file arts_2023_model_coefficients.csv; Sample summary is backed by the local file arts_2023_sample_summary.csv; Fit summary is backed by the local file arts_2023_model_fit.csv. The resulting executable panel supplies 5,192 observations across 443 entities and covers 1991 to 2015 in the local executable panel.. Every numeric claim in the results section is intended to trace back to one of those verified files rather than to a hand-copied number from the article. This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct

cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated. The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. The local sample frame is Processed public-firm GVKEY-year panel with patent portfolios and financial controls. The observed time window in the executable panel is 1991 to 2015 in the local executable panel. The outcome is Winsorized Tobin's Q at the firm-year level. The main executable panel used in the paper is arts_2023_analysis_dataset.csv and the principal coefficient export is arts_2023_model_coefficients.csv .

File	Role
arts_2023_analysis_dataset.csv	Firm-year analysis panel
arts_2023_model_coefficients.csv	Estimated coefficients
arts_2023_model_fit.csv	Model fit statistics
arts_2023_sample_summary.csv	Sample reconstruction summary

Core local datasets and result files used in the current paper.

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Arts, Cassiman, and Hou (2023). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument. The analysis panel used in this paper is arts_2023_analysis_dataset.csv. Its role is to estimate the local model and regenerate descriptive statistics. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on

local machine details. The coefficient export used in this paper is arts_2023_model_coefficients.csv. Its role is to recover the reported local coefficients and p-values. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. The sample summary used in this paper is arts_2023_sample_summary.csv. Its role is to audit sample construction and raw grouped moments. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. The fit summary used in this paper is arts_2023_model_fit.csv. Its role is to check model diagnostics when a fit file is available. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim.

Stage	NeededFor	Dataset	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	arts_2023_analysis_dataset.csv	Yes	Executable local dataset

Stage	NeededFor	Dataset	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	arts_2023_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	arts_2023_sample_summary.csv	Yes	Executable local result file
Fit summary	Check model diagnostics when a fit file is available	arts_2023_model_fit.csv	Yes	Executable local result file

Verified current files used to support the paper's calculations.

3.2 Sample Reconstruction

The local analysis file contains 5,192 observations drawn from 443 entities. The observed time span runs from 1,991 to 2,015. Selected sample moments are tobin_q_w = 2.1771, tech_diff = 0.6682 . These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed.

Statistic	Value
rows	5192.0000
unique_firms	443.0000
year_min	1991.0000
year_max	2015.0000
mean_tobin_q_w	2.1771
mean_tech_diff	0.6682
sd_tech_diff	0.1733

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Technology portfolio is operationalized locally as Five-year rolling primary-USPC portfolio and appears in the executable dataset as portfolio_patents; class_count; class_hhi. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Technology differentiation is operationalized locally as One minus peer similarity and appears in the executable dataset as tech_diff. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Compute one minus average similarity to close peers.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Peer benchmark is operationalized locally as Nearest peers in year t and appears in the executable dataset as neighbor_similarity. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use local class-based portfolios to define the similarity set.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is

that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Firm performance is operationalized locally as Winsorized Tobin's Q and appears in the executable dataset as tobin_q_w. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reconstruct Tobin's Q from processed Compustat variables and winsorize.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Financial controls is operationalized locally as Assets, leverage, cash, R&D, portfolio size and appears in the executable dataset as log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents). The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the financial controls already built in the local script.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Technology portfolio	Five-year rolling primary-USPC portfolio	portfolio_patents; class_count; class_hhi	Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.
Technology differentiation	One minus peer similarity	tech_diff	Compute one minus average similarity to close peers.
Peer benchmark	Nearest peers in year t	neighbor_similarity	Use local class-based portfolios to define the similarity set.
Firm performance	Winsorized Tobin's Q	tobin_q_w	Reconstruct Tobin's Q from processed Compustat variables and winsorize.
Financial controls	Assets, leverage, cash, R&D, portfolio size	log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents)	Carry forward the financial controls already built in the local script.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Arts, Cassiman, and Hou (2023). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Technology portfolio, the manuscript uses Five-year rolling primary-USPC portfolio under the exact local implementation portfolio_patents; class_count; class_hhi. The derivation step is: Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Technology differentiation, the manuscript uses One minus peer similarity under the exact local implementation tech_diff. The derivation step is: Compute one minus average similarity to close peers.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Peer benchmark, the manuscript uses Nearest peers in year t under the exact local implementation neighbor_similarity. The derivation step is: Use local class-based portfolios to define the similarity set.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm performance, the manuscript uses Winsorized Tobin's Q under the exact local implementation tobin_q_w. The derivation step is: Reconstruct Tobin's Q from processed Compustat variables and winsorize.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Financial controls, the manuscript uses Assets, leverage, cash, R&D, portfolio size under the exact local implementation log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents). The derivation step is: Carry forward the financial controls already built in the local script.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Winsorized Tobin's Q has mean 2.1771, standard deviation 1.5081, and an interquartile range from

1.3189 to 2.4637; Technology differentiation has mean 0.6682, standard deviation 0.1733, and an interquartile range from 0.5576 to 0.8000; Neighbor similarity has mean 0.3318, standard deviation 0.1733, and an interquartile range from 0.2000 to 0.4424; Portfolio patents has mean 552.7049, standard deviation 1483.5605, and an interquartile range from 29.0000 to 450.0000 . The time-series view is equally informative. V1 changes from 233.0000 in the first observed period to 204.0000 in the last observed period; V2 changes from 0.6696 in the first observed period to 0.6297 in the last observed period; V3 changes from 1.7221 in the first observed period to 2.4654 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output. The regressor-quantile analysis shows that mean outcomes range from 1.9569 to 2.8572 across the support of the focal regressor, while the mean regressor itself ranges from 0.4011 to 0.8832. That pattern is useful because it shows what the formal model is trying to account for in the current dataset. The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is.

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. R&D intensity has missing share 9.77; Winsorized Tobin's Q has missing share 0.00; Technology differentiation has missing share 0.00 . At the other end of the support, Winsorized Tobin's Q remains close to complete with missing share 0.00; Technology differentiation remains close to complete with missing share 0.00; Neighbor similarity remains close to complete with missing share 0.00 . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively. Coverage across entities is also visible in the current panel rather than assumed. 10016 contributes 23 observations; 10405 contributes 23 observations; 10499 contributes 23 observations; 10519 contributes 23 observations; 1078 contributes 23 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow

set of heavily observed entities. The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 0.8832 and the mean outcome is 1.9569 across 1,039 observations. In the highest observed quantile, the corresponding values are 0.4011, 2.8572, and 1,039 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells. These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because it tests whether a central strategy result about technological differentiation survives when the semantic layer is stripped back to a fully executable local panel. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication.

Variable	Missing	MissingShare
Winsorized Tobin's Q	0.0000	0.0000
Technology differentiation	0.0000	0.0000
Neighbor similarity	0.0000	0.0000
Portfolio patents	0.0000	0.0000
Class count	0.0000	0.0000
Class concentration	0.0000	0.0000
Log assets	0.0000	0.0000
Leverage	0.0000	0.0000
Cash ratio	0.0000	0.0000
R&D intensity	507.0000	0.0977

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Winsorized Tobin's Q	5192.0000	2.1771	1.5081	1.3189	1.7253	2.4637	0.4751	23.07
Technology differentiation	5192.0000	0.6682	0.1733	0.5576	0.6935	0.8000	0.0866	0.99
Neighbor similarity	5192.0000	0.3318	0.1733	0.2000	0.3065	0.4424	0.0050	0.91
Portfolio patents	5192.0000	552.7049	1483.5605	29.0000	129.0000	450.0000	1.0000	33760.0
Class count	5192.0000	41.7427	42.9175	10.0000	28.0000	57.0000	1.0000	250.00
Class concentration	5192.0000	0.2126	0.2196	0.0714	0.1347	0.2653	0.0163	1.00
Log assets	5192.0000	8.8404	1.3569	7.9058	8.7190	9.6639	4.3114	13.58
Leverage	5192.0000	0.5877	0.2216	0.4541	0.5836	0.7112	0.0317	2.93
Cash ratio	5192.0000	0.1348	0.1468	0.0280	0.0802	0.1909	0.0000	0.85
R&D intensity	4685.0000	0.0519	0.0540	0.0146	0.0327	0.0741	0.0000	0.67

Main descriptive statistics discussed in the body of the paper.

Period	V1	V2	V3	V4
1991.0000	233.0000	0.6696	1.7221	320.5064
1992.0000	237.0000	0.6653	1.7417	325.8270
1993.0000	238.0000	0.6760	1.7913	337.9874
1994.0000	239.0000	0.6778	1.7348	349.7071
1995.0000	236.0000	0.6830	2.0166	366.8051
1996.0000	236.0000	0.6594	2.1379	380.8602
1997.0000	233.0000	0.6489	2.4754	404.3605
1998.0000	221.0000	0.6501	2.6527	440.2896
1999.0000	211.0000	0.6509	2.9393	488.1280
2000.0000	221.0000	0.6796	2.9787	519.8326

Period	V1	V2	V3	V4
2001.0000	238.0000	0.6826	2.4734	543.5042
2002.0000	237.0000	0.6895	1.9995	526.8692
2005.0000	227.0000	0.6956	2.2091	268.2907
2006.0000	220.0000	0.6876	2.2944	280.4364
2007.0000	210.0000	0.6890	2.3005	374.0714
2008.0000	214.0000	0.6862	1.7092	516.5421
2009.0000	225.0000	0.6863	1.8849	640.5956
2010.0000	227.0000	0.6709	2.0194	736.5771
2011.0000	225.0000	0.6578	1.9323	811.2578
2012.0000	220.0000	0.6524	1.9836	890.6409
2013.0000	223.0000	0.6351	2.2694	974.1121
2014.0000	217.0000	0.6389	2.5169	1106.1613
2015.0000	204.0000	0.6297	2.4654	1249.4461

Observed time patterns in the current dataset.

Quantile	N	MeanRegressor	MeanOutcome
5.0000	1039.0000	0.8832	1.9569
4.0000	1038.0000	0.7780	1.9779
3.0000	1038.0000	0.6925	1.9966
2.0000	1038.0000	0.5863	2.0965
1.0000	1039.0000	0.4011	2.8572

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Winsorized Tobin's Q	Technology differentiation	Neighbor similarity	Portfolio patents	Class count	Class concentration	Log assets	Leverage	Cash ratio
Winsorized Tobin's Q	1.000	-0.214	0.214	-0.029	-0.144	0.213	-0.097	-0.245	0.357
Technology differentiation	-0.214	1.000	-1.000	-0.221	-0.153	0.106	-0.224	0.125	-0.298
Neighbor similarity	0.214	-1.000	1.000	0.221	0.153	-0.106	0.224	-0.125	0.298
Portfolio patents	-0.029	-0.221	0.221	1.000	0.627	-0.200	0.414	0.037	0.044
Class count	-0.144	-0.153	0.153	0.627	1.000	-0.503	0.561	0.151	-0.124
Class concentration	0.213	0.106	-0.106	-0.200	-0.503	1.000	-0.249	-0.110	0.130

Compact correlation structure for the main variables.

Entity	Observations	FirstPeriod	LastPeriod
10016.0000	23.0000	1991.0000	2015.0000
10405.0000	23.0000	1991.0000	2015.0000
10499.0000	23.0000	1991.0000	2015.0000
10519.0000	23.0000	1991.0000	2015.0000
1078.0000	23.0000	1991.0000	2015.0000
10983.0000	23.0000	1991.0000	2015.0000
10984.0000	23.0000	1991.0000	2015.0000
11456.0000	23.0000	1991.0000	2015.0000
11465.0000	23.0000	1991.0000	2015.0000
1161.0000	23.0000	1991.0000	2015.0000
11636.0000	23.0000	1991.0000	2015.0000

Entity	Observations	FirstPeriod	LastPeriod
1209.0000	23.0000	1991.0000	2015.0000
1300.0000	23.0000	1991.0000	2015.0000
1356.0000	23.0000	1991.0000	2015.0000
1690.0000	23.0000	1991.0000	2015.0000
1722.0000	23.0000	1991.0000	2015.0000
1913.0000	23.0000	1991.0000	2015.0000
1988.0000	23.0000	1991.0000	2015.0000
2044.0000	23.0000	1991.0000	2015.0000
2086.0000	23.0000	1991.0000	2015.0000

Most observed entities in the current dataset.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism. Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest. The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Processed public-firm GVKEY-year panel with patent portfolios and financial controls., and the empirical window is 1991 to 2015 in the local executable panel.. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample. The second set of decisions concerns measurement and specification. The outcome is Winsorized Tobin's Q at the firm-year level., and the main explanatory design is Firm and year fixed-effects regression of Tobin's

Q on local technology differentiation and controls.. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question. The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Financial controls plus firm and year fixed effects.. The crucial exclusion or substitution rule is: Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability. A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether technologically differentiated firms perform better than their peers.
Unit of analysis	Firm-year
Sample frame	Processed public-firm GVKEY-year panel with patent portfolios and financial controls.
Time window	1991 to 2015 in the local executable panel.
Outcome definition	Winsorized Tobin's Q at the firm-year level.

DecisionPoint	LocalSpecification
Main explanatory design	Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls.
Controls and fixed effects	Financial controls plus firm and year fixed effects.
Exclusion or substitution logic	Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Processed public-firm GVKEY-year panel with patent portfolios and financial controls. over 1991 to 2015 in the local executable panel.. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Winsorized Tobin's Q at the firm-year level. and the main explanatory design is Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls.. The local paper keeps these objects

because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Financial controls plus firm and year fixed effects. The main exclusion or substitution rule is Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$Q_{it}^{(w)} = \beta_1 \text{TechDiff}_{it} + \beta_2 X_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.1935 on tech_diff (p = 0.7841). The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. Log assets is estimated at -0.5161 with p = 0.0017. Log(1 + portfolio patents) is estimated at -0.2660 with p = 0.0021. R&D intensity is estimated at 4.2903 with p = 0.0093. The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

model	Term	estimate	std_error	p_value
arts_fe	Technology differentiation	-0.1935	0.6974	0.7841
arts_fe	Log assets	-0.5161	0.1442	0.0017
arts_fe	Leverage	-0.6522	0.2840	0.0315
arts_fe	Cash ratio	1.5652	0.6384	0.0226
arts_fe	R&D intensity	4.2903	1.5054	0.0093

model	Term	estimate	std_error	p_value
arts_fe	Log(1 + portfolio patents)	-0.2660	0.0764	0.0021
arts_fe	Class count	0.0043	0.0024	0.0851
arts_fe	Class concentration	0.4918	0.2947	0.1094

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log assets is estimated at -0.5161 with a p-value of 0.0017; Log(1 + portfolio patents) is estimated at -0.2660 with a p-value of 0.0021; R&D intensity is estimated at 4.2903 with a p-value of 0.0093; Cash ratio is estimated at 1.5652 with a p-value of 0.0226 . Within the focal specification, the sharpest local signal comes from Log assets, which is estimated at -0.5161 with p = 0.0017. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment. The scale of the underlying variables also matters for interpretation. The outcome Winsorized Tobin's Q has mean 2.1771 and standard deviation 1.5081 in the local panel, while the main regressor Technology differentiation has mean 0.6682 and standard deviation 0.1733. A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 1.9569 to 2.8572 while the mean regressor ranges from 0.4011 to 0.8832. This does not substitute for the model, but it shows how much raw variation the specification must explain. Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation.

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification. Within arts_fe, the leading coefficient family reads as follows: Log assets = -0.5161 (p = 0.0017); Log(1 + portfolio patents) = -0.2660 (p = 0.0021); R&D intensity = 4.2903 (p = 0.0093); Cash ratio = 1.5652 (p = 0.0226); Leverage = -0.6522 (p = 0.0315). The largest positive estimate in this specification is R&D intensity at 4.2903, while the largest negative estimate is Leverage at -0.6522. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. This fuller coefficient reading is especially important for Arts, Cassiman, and Hou (2023) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text.

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In arts_fe, the most informative focal term is Log assets with estimate -0.5161 and p = 0.0017 . The arts_fe specification contributes its own reading to the replication. Log assets is estimated at -0.5161 with p = 0.0017 and is classified as a strong signal; Log(1 + portfolio patents) is estimated at -0.2660 with p = 0.0021 and is classified as a strong signal; R&D intensity is estimated at 4.2903 with p = 0.0093 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. This model comparison matters for Arts, Cassiman, and Hou (2023) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes

sign under a different outcome definition.

Term	Model	Estimate	PValue	Signal	Reading
Technology differentiation	arts_fe	-0.1935	0.7841	Weak	Negative estimate with p = 0.7841.
Log assets	arts_fe	-0.5161	0.0017	Strong	Negative estimate with p = 0.0017.
Leverage	arts_fe	-0.6522	0.0315	Moderate	Negative estimate with p = 0.0315.
Cash ratio	arts_fe	1.5652	0.0226	Moderate	Positive estimate with p = 0.0226.
R&D intensity	arts_fe	4.2903	0.0093	Strong	Positive estimate with p = 0.0093.
Log(1 + portfolio patents)	arts_fe	-0.2660	0.0021	Strong	Negative estimate with p = 0.0021.

Term	Model	Estimate	PValue	Signal	Reading
Class count	arts_fe	0.0043	0.0851	Suggestive	Positive estimate with p = 0.0851.
Class concentration	arts_fe	0.4918	0.1094	Weak	Positive estimate with p = 0.1094.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation. For arts_fe, the current fit export reports N = 4,669, log-likelihood = -6268.8113, AIC = 13389.6226, BIC = 16136.7689, within R-squared = 0.1440. These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Arts, Cassiman, and Hou (2023) is being carried out on a clearly described estimating sample with observable fit statistics.

n	ll	aic	bic	wr2	model
4669.0000	-6268.8113	13389.6226	16136.7689	0.1440	arts_fe

Model fit statistics discussed in the main results section.

term	estimate	std_error	statistic	p_value	model
tech_diff	-0.1935	0.6974	-0.2774	0.7841	arts_fe
log_assets	-0.5161	0.1442	-3.5785	0.0017	arts_fe
leverage	-0.6522	0.2840	-2.2967	0.0315	arts_fe
cash_ratio	1.5652	0.6384	2.4517	0.0226	arts_fe
rd_intensity	4.2903	1.5054	2.8500	0.0093	arts_fe
log1p(portfolio_patents)	-0.2660	0.0764	-3.4800	0.0021	arts_fe
class_count	0.0043	0.0024	1.8033	0.0851	arts_fe
class_hhi	0.4918	0.2947	1.6685	0.1094	arts_fe

Full coefficient table from the current local estimation output.

7 Comparison to the Published Study

The comparison to the published paper is intentionally narrow. The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication. This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation.

8 Discussion

The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly. The first limitation concerns measurement. The exact text-embedding pipeline from the source paper is not stored locally. The second limitation concerns sample translation. The local differentiation measure is therefore class-based rather than text-semantic. The third limitation concerns interpretation. Comparisons to the published estimates should be read as inferential rather than byte-identical.

9 Conclusion

The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit. The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available.

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value
rows	5192.0000
unique_firms	443.0000
year_min	1991.0000
year_max	2015.0000
mean_tobin_q_w	2.1771
mean_tech_diff	0.6682
sd_tech_diff	0.1733

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Winsorized Tobin's Q	5192.0000	2.1771	1.5081	1.3189	1.7253	2.4637	0.4751	23.07
Technology differentiation	5192.0000	0.6682	0.1733	0.5576	0.6935	0.8000	0.0866	0.99
Neighbor similarity	5192.0000	0.3318	0.1733	0.2000	0.3065	0.4424	0.0050	0.91
Portfolio patents	5192.0000	552.7049	1483.5605	29.0000	129.0000	450.0000	1.0000	33760.0
Class count	5192.0000	41.7427	42.9175	10.0000	28.0000	57.0000	1.0000	250.00
Class concentration	5192.0000	0.2126	0.2196	0.0714	0.1347	0.2653	0.0163	1.00
Log assets	5192.0000	8.8404	1.3569	7.9058	8.7190	9.6639	4.3114	13.58
Leverage	5192.0000	0.5877	0.2216	0.4541	0.5836	0.7112	0.0317	2.93
Cash ratio	5192.0000	0.1348	0.1468	0.0280	0.0802	0.1909	0.0000	0.85
R&D intensity	4685.0000	0.0519	0.0540	0.0146	0.0327	0.0741	0.0000	0.67

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Winsorized Tobin's Q	Technology differentiation	Neighbor similarity	Portfolio patents	Class count	Class concentration	Log assets	Leverage	Cash ratio
Winsorized Tobin's Q	1.000	-0.214	0.214	-0.029	-0.144	0.213	-0.097	-0.245	0.357
Technology differentiation	-0.214	1.000	-1.000	-0.221	-0.153	0.106	-0.224	0.125	-0.298
Neighbor similarity	0.214	-1.000	1.000	0.221	0.153	-0.106	0.224	-0.125	0.298
Portfolio patents	-0.029	-0.221	0.221	1.000	0.627	-0.200	0.414	0.037	0.044
Class count	-0.144	-0.153	0.153	0.627	1.000	-0.503	0.561	0.151	-0.124
Class concentration	0.213	0.106	-0.106	-0.200	-0.503	1.000	-0.249	-0.110	0.130
Log assets	-0.097	-0.224	0.224	0.414	0.561	-0.249	1.000	0.140	-0.133
Leverage	-0.245	0.125	-0.125	0.037	0.151	-0.110	0.140	1.000	-0.364
Cash ratio	0.357	-0.298	0.298	0.044	-0.124	0.130	-0.133	-0.364	1.000
R&D intensity	0.340	-0.396	0.396	0.053	-0.051	0.112	-0.205	-0.315	0.513

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	V1	V2	V3	V4
1991.0000	233.0000	0.6696	1.7221	320.5064
1992.0000	237.0000	0.6653	1.7417	325.8270
1993.0000	238.0000	0.6760	1.7913	337.9874
1994.0000	239.0000	0.6778	1.7348	349.7071
1995.0000	236.0000	0.6830	2.0166	366.8051
1996.0000	236.0000	0.6594	2.1379	380.8602
1997.0000	233.0000	0.6489	2.4754	404.3605
1998.0000	221.0000	0.6501	2.6527	440.2896
1999.0000	211.0000	0.6509	2.9393	488.1280
2000.0000	221.0000	0.6796	2.9787	519.8326
2001.0000	238.0000	0.6826	2.4734	543.5042
2002.0000	237.0000	0.6895	1.9995	526.8692
2005.0000	227.0000	0.6956	2.2091	268.2907
2006.0000	220.0000	0.6876	2.2944	280.4364
2007.0000	210.0000	0.6890	2.3005	374.0714
2008.0000	214.0000	0.6862	1.7092	516.5421
2009.0000	225.0000	0.6863	1.8849	640.5956
2010.0000	227.0000	0.6709	2.0194	736.5771
2011.0000	225.0000	0.6578	1.9323	811.2578
2012.0000	220.0000	0.6524	1.9836	890.6409
2013.0000	223.0000	0.6351	2.2694	974.1121
2014.0000	217.0000	0.6389	2.5169	1106.1613
2015.0000	204.0000	0.6297	2.4654	1249.4461

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
5.0000	1039.0000	0.8832	1.9569
4.0000	1038.0000	0.7780	1.9779
3.0000	1038.0000	0.6925	1.9966
2.0000	1038.0000	0.5863	2.0965
1.0000	1039.0000	0.4011	2.8572

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
tech_diff	-0.1935	0.6974	-0.2774	0.7841	arts_fe
log_assets	-0.5161	0.1442	-3.5785	0.0017	arts_fe
leverage	-0.6522	0.2840	-2.2967	0.0315	arts_fe
cash_ratio	1.5652	0.6384	2.4517	0.0226	arts_fe
rd_intensity	4.2903	1.5054	2.8500	0.0093	arts_fe
log1p(portfolio_patents)	-0.2660	0.0764	-3.4800	0.0021	arts_fe
class_count	0.0043	0.0024	1.8033	0.0851	arts_fe
class_hhi	0.4918	0.2947	1.6685	0.1094	arts_fe

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Winsorized Tobin's Q	0.0000	0.0000
Technology differentiation	0.0000	0.0000
Neighbor similarity	0.0000	0.0000
Portfolio patents	0.0000	0.0000
Class count	0.0000	0.0000
Class concentration	0.0000	0.0000
Log assets	0.0000	0.0000
Leverage	0.0000	0.0000
Cash ratio	0.0000	0.0000
R&D intensity	507.0000	0.0977

Missingness audit for the variables summarized in the manuscript.

Entity	Observations	FirstPeriod	LastPeriod
10016.0000	23.0000	1991.0000	2015.0000
10405.0000	23.0000	1991.0000	2015.0000
10499.0000	23.0000	1991.0000	2015.0000
10519.0000	23.0000	1991.0000	2015.0000
1078.0000	23.0000	1991.0000	2015.0000
10983.0000	23.0000	1991.0000	2015.0000
10984.0000	23.0000	1991.0000	2015.0000
11456.0000	23.0000	1991.0000	2015.0000
11465.0000	23.0000	1991.0000	2015.0000
1161.0000	23.0000	1991.0000	2015.0000
11636.0000	23.0000	1991.0000	2015.0000
1209.0000	23.0000	1991.0000	2015.0000

Entity	Observations	FirstPeriod	LastPeriod
1300.0000	23.0000	1991.0000	2015.0000
1356.0000	23.0000	1991.0000	2015.0000
1690.0000	23.0000	1991.0000	2015.0000
1722.0000	23.0000	1991.0000	2015.0000
1913.0000	23.0000	1991.0000	2015.0000
1988.0000	23.0000	1991.0000	2015.0000
2044.0000	23.0000	1991.0000	2015.0000
2086.0000	23.0000	1991.0000	2015.0000

Top entities by local observation count.

References

[1] Sam Arts, Bruno Cassiman, and Jianan Hou. Position and differentiation of firms in technology space. *Management Science*, 2023.